PyNoxtli

Version 0.1

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BVP in the whole domain

$$\mathcal{L}u = f_\Omega$$

$$\underline{\underline{M}}\,\underline{U} = \underline{F}$$

subdom1.png

Finite Volume Method

meshFVM.pdf

- The variables are calculated on the centers of the volumes.
- Discretized equations in 2D:

$$\begin{aligned} \boldsymbol{a_P}\psi_{\boldsymbol{P}} &= \boldsymbol{a_E}\psi_{\boldsymbol{E}} + \boldsymbol{a_W}\psi_{\boldsymbol{W}} + \\ &\quad \boldsymbol{a_N}\psi_{\boldsymbol{N}} + \boldsymbol{a_S}\psi_{\boldsymbol{S}} + S_{\boldsymbol{P}} \end{aligned}$$

- Discretized equations in 3D:

$$\begin{aligned} \boldsymbol{a_P}\psi_{\boldsymbol{P}} &= \boldsymbol{a_E}\psi_{\boldsymbol{E}} + \boldsymbol{a_W}\psi_{\boldsymbol{W}} + \\ &\quad \boldsymbol{a_N}\psi_{\boldsymbol{N}} + \boldsymbol{a_S}\psi_{\boldsymbol{S}} + \\ &\quad \boldsymbol{a_F}\psi_{\boldsymbol{F}} + \boldsymbol{a_B}\psi_{\boldsymbol{B}} + S_{\boldsymbol{P}} \end{aligned}$$

- $\boldsymbol{P} = i,j; \boldsymbol{E} = i+1,j; \boldsymbol{W} = i-1,j; \boldsymbol{N} = i,j+1; \boldsymbol{S} = i,j-1$
- FVM equations: $\boxed{\boldsymbol{a_{i,j}}\psi_{\boldsymbol{i,j}} = \boldsymbol{a_{i+1,j}}\psi_{\boldsymbol{i+1,j}} + \boldsymbol{a_{i-1,j}}\psi_{\boldsymbol{i-1,j}} + \boldsymbol{a_{i,j+1}}\psi_{\boldsymbol{i,j+1}} + \boldsymbol{a_{i,j-1}}\psi_{\boldsymbol{i,j-1}} + S_{\boldsymbol{i,j}}}$

Finite Volume Method: Linear Systems 2D

tdma2D.jpg

Finite Volume Method: Linear Systems 3D

tdma3D.jpg

Finite Volume Method: example in 2D

meshFVM.pdf

Finite Volume Method: matrix

i		1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
	j			1					2					3					4		
1	1	P	E				N														
2	1	W	P	E				N													
3	1		W	P	E				N												
4	1			W	P	E				N											
5	1				W	P					N										
1	2	S					P	E				N									
2	2		S				W	P	E				N								
3	2			S				W	P	E				N							
4	2				S				W	P	E				N						
5	2					S				W	P					N					
1	3						S					P	E				N				
2	3							S				W	P	E				N			
3	3								S				W	P	E				N		
4	3									S				W	P	E				N	
5	3										S				W	P					N
1	4											S					P	E			
2	4												S				W	P	E		
3	4													S				W	P	E	
4	4														S				W	W	E
5	4															S				W	P

$$a_P = a_E + a_W + a_N + a_S$$

Original domain and nodes (*Fine mesh*)

subdom1.png

Original domain partitioned into subdomains

subdom2.png

Splitting of nodes : **Derived Nodes**

Non-overlapping formulation: $\boxed{\underline{\underline{a}}\,\underline{\underline{A}}^t\,\underline{u} = \underline{f} \;\text{ and }\; \underline{\underline{j}}\,\underline{u} = 0}$

subdom5.png

Derived nodes and its multiplicity

Non-overlapping formulation: $\boxed{\underline{\underline{a}}\,\underline{\underline{A}}^t\,\underline{u} = \underline{f} \;\text{ and }\; \underline{\underline{j}}\,\underline{u} = 0}$ $\quad \underline{\underline{A}}^t = \sum_{\gamma=1}^{E} \underline{\underline{A}}^{\gamma}$

subdom3.png

Identification of the derived nodes

I : Interior nodes $\quad \Pi = I \cup \pi \quad \Sigma = I \cup \Delta$

Δ : Dual nodes $\quad X = \Pi \cup \Delta$: Derived nodes

π : Primal nodes $\quad \Gamma$: Interior boundaries

All the operations of DVS algorithms are done locally in each subdomain

subdom4.png

Matrices

$$\underline{\underline{A}}^t = \begin{pmatrix} \underline{\underline{A}}^t_{II} & \underline{\underline{A}}^t_{I\Delta} & \underline{\underline{A}}^t_{I\pi} \\ \underline{\underline{A}}^t_{\Delta I} & \underline{\underline{A}}^t_{\Delta\Delta} & \underline{\underline{A}}^t_{\Delta\pi} \\ \underline{\underline{A}}^t_{\pi I} & \underline{\underline{A}}^t_{\pi\Delta} & \underline{\underline{A}}^t_{\pi\pi} \end{pmatrix} \equiv \begin{pmatrix} \underline{\underline{A}}^t_{\Pi\Pi} & \underline{\underline{A}}^t_{\Pi\Delta} \\ \underline{\underline{A}}^t_{\Delta\Pi} & \underline{\underline{A}}^t_{\Delta\Delta} \end{pmatrix}$$

$$\underline{\underline{A}}^t_{\Pi\Pi} = \begin{pmatrix} \underline{\underline{A}}^t_{II} & \underline{\underline{A}}^t_{I\pi} \\ \underline{\underline{A}}^t_{\pi I} & \underline{\underline{A}}^t_{\pi\pi} \end{pmatrix}, \qquad \underline{\underline{A}}^t_{\Pi\Delta} = \begin{pmatrix} \underline{\underline{A}}^t_{I\Delta} \\ \underline{\underline{A}}^t_{\pi\Delta} \end{pmatrix}$$

$$\underline{\underline{A}}^t_{\Delta\Pi} = \begin{pmatrix} \underline{\underline{A}}^t_{\Delta I} & \underline{\underline{A}}^t_{\Delta\pi} \end{pmatrix}, \qquad \underline{\underline{A}}^t_{\Delta\Delta} = \begin{pmatrix} \underline{\underline{A}}^t_{\Delta\Delta} \end{pmatrix}$$

Matrices

$$\underline{\underline{A}}^t = \begin{pmatrix} \underline{\underline{A}}^t_{II} & \underline{\underline{A}}^t_{I\Delta} & \underline{\underline{A}}^t_{I\pi} \\ \underline{\underline{A}}^t_{\Delta I} & \underline{\underline{A}}^t_{\Delta\Delta} & \underline{\underline{A}}^t_{\Delta\pi} \\ \underline{\underline{A}}^t_{\pi I} & \underline{\underline{A}}^t_{\pi\Delta} & \underline{\underline{A}}^t_{\pi\pi} \end{pmatrix} \equiv \begin{pmatrix} \underline{\underline{A}}^t_{\Pi\Pi} & \underline{\underline{A}}^t_{\Pi\Delta} \\ \underline{\underline{A}}^t_{\Delta\Pi} & \underline{\underline{A}}^t_{\Delta\Delta} \end{pmatrix}$$

$$\underline{\underline{A}}^t_{\Pi\Pi} = \begin{pmatrix} \underline{\underline{A}}^t_{II} & \underline{\underline{A}}^t_{I\pi} \\ \underline{\underline{A}}^t_{\pi I} & \underline{\underline{A}}^t_{\pi\pi} \end{pmatrix}, \qquad \underline{\underline{A}}^t_{\Pi\Delta} = \begin{pmatrix} \underline{\underline{A}}^t_{I\Delta} \\ \underline{\underline{A}}^t_{\pi\Delta} \end{pmatrix}$$

$$\underline{\underline{A}}^t_{\Delta\Pi} = \begin{pmatrix} \underline{\underline{A}}^t_{\Delta I} & \underline{\underline{A}}^t_{\Delta\pi} \end{pmatrix}, \qquad \underline{\underline{A}}^t_{\Delta\Delta} = \begin{pmatrix} \underline{\underline{A}}^t_{\Delta\Delta} \end{pmatrix}$$

The Preconditioned DVS – Algorithms

Four algorithms

Name	Operator	Constraint
DVS-BDDC	$\underline{\underline{a}}\,\underline{\underline{S}}^{-1}\,\underline{\underline{a}}\,\underline{\underline{S}}\,\underline{u}_{\Delta} = \underline{\underline{a}}\,\underline{\underline{S}}^{-1}\,\underline{f}_{\Delta}$	$\underline{\underline{j}}\,\underline{u}_{\Delta} = 0$
DVS-FETI-DP	$\underline{\underline{j}}\,\underline{\underline{S}}\,\underline{\underline{j}}\,\underline{\underline{S}}^{-1}\,\underline{\lambda} = \underline{\underline{j}}\,\underline{\underline{S}}\,\underline{\underline{j}}\,\underline{\underline{S}}^{-1}\,\underline{f}_{\Delta}$	$\underline{\underline{a}}\,\underline{\lambda} = 0$
DVS-PRIMAL	$\underline{\underline{S}}^{-1}\,\underline{\underline{j}}\,\underline{\underline{S}}\,\underline{\underline{j}}\,\underline{\mathrm{v}}_{\Delta} = \underline{\underline{S}}^{-1}\,\underline{\underline{j}}\,\underline{\underline{S}}\,\underline{\underline{j}}\,\underline{\underline{S}}^{-1}\,\underline{f}_{\Delta}$	$\underline{\underline{a}}\,\underline{\underline{S}}\,\underline{\mathrm{v}}_{\Delta} = 0$
DVS-DUAL	$\underline{\underline{S}}\,\underline{\underline{a}}\,\underline{\underline{S}}^{-1}\,\underline{\underline{a}}\,\underline{\mu} = \underline{\underline{S}}\,\underline{\underline{a}}\,\underline{\underline{S}}^{-1}\,\underline{\underline{a}}\,\underline{\underline{S}}\,\underline{\underline{j}}\,\underline{\underline{S}}^{-1}\,\underline{f}_{\Delta}$	$\underline{\underline{j}}\,\underline{\underline{S}}^{-1}\,\underline{\mu} = 0$

DVS–BDDC

- DVS-algorithms can be posed in the form:
  - $\underline{\underline{M}}\,\underline{u}_\Delta = \underline{b}_\Delta$ with restriction $\underline{\underline{R}}\,\underline{u}_\Delta$
- In the case of DVS-BDDC we have:
  - $\underline{\underline{M}} = \underline{\underline{a}}\,\underline{\underline{S}}^{-1}\,\underline{\underline{a}}\,\underline{\underline{S}}$
  - $\underline{b}_\Delta = \underline{\underline{a}}\,\underline{\underline{S}}^{-1}\,\underline{f}_\Delta$
  - $\underline{\underline{R}} = \underline{\underline{j}}$

DVS-BDDC in action

$$\boxed{\underline{\underline{M}}\,\underline{u}_\Delta = \underline{\underline{a}}\,\underline{\underline{S}}^{-1}\,\underline{\underline{a}}\,\underline{\underline{S}}\,\underline{u}_\Delta}$$

1. Apply Schur: $\underline{vec}_\Delta \leftarrow \underline{\underline{S}}\,\underline{u}_\Delta$
2. Apply Average: $\underline{vec}_\Delta \leftarrow \underline{\underline{a}}\,\underline{vec}_\Delta$
3. Apply Inverse of Schur: $\underline{vec}_\Delta \leftarrow \underline{\underline{S}}^{-1}\,\underline{vec}_\Delta$
4. Apply Average: $\underline{vec}_\Delta \leftarrow \underline{\underline{a}}\,\underline{vec}_\Delta$

$$\boxed{\underline{b}_\Delta = \underline{\underline{a}}\,\underline{\underline{S}}^{-1}\,\underline{f}_\Delta}$$

1. Apply Inverse of Schur: $\underline{vec}_\Delta \leftarrow \underline{\underline{S}}^{-1}\,\underline{f}_\Delta$
2. Apply Average: $\underline{vec}_\Delta \leftarrow \underline{\underline{a}}\,\underline{vec}_\Delta$

Matrix Notations

- All matrices are linear transformations of W into W: $\underline{\underline{B}} : W \to W$
- Let $M \in X$, $N \in X$ and $M \cap N = \varnothing$ subsets of X

domain06.png domain07.png domain08.png

We write:

$$\underline{\underline{B}} = \begin{pmatrix} \underline{\underline{B}}_{MM} & \underline{\underline{B}}_{MN} \\ \underline{\underline{B}}_{NM} & \underline{\underline{B}}_{NN} \end{pmatrix} \qquad \underline{\underline{B}}^{-1} = \begin{pmatrix} (\underline{\underline{B}}^{-1})_{MM} & (\underline{\underline{B}}^{-1})_{MN} \\ (\underline{\underline{B}}^{-1})_{NM} & (\underline{\underline{B}}^{-1})_{NN} \end{pmatrix}$$

General Schur Formulas

1. $$\boxed{\sigma_{NN}(\underline{\underline{B}})\underline{\mathrm{v}}_N = \underline{g}_N}$$
   - $\sigma_{NN}(\underline{\underline{B}}) = \underline{\underline{B}}_{NN} - \underline{\underline{B}}_{NM}(\underline{\underline{B}}_{MM})^{-1}\underline{\underline{B}}_{MN}$
   - $\underline{g}_N = \underline{w}_N - \underline{\underline{B}}_{NM}(\underline{\underline{B}}_{MM})^{-1}\underline{w}_M$
2. $$\boxed{\underline{\mathrm{v}}_M = (\underline{\underline{B}}_{MM})^{-1}(\underline{w}_M - \underline{\underline{B}}_{MN}\underline{\mathrm{v}}_N)}$$
   - $\underline{\underline{B}}_{MM}\underline{\mathrm{v}}_M = (\underline{w}_M - \underline{\underline{B}}_{MN}\underline{\mathrm{v}}_N)$

Application of $\underline{\underline{S}}$

$$\underline{\underline{S}} \equiv \underline{\underline{A}}_{\Delta\Delta} - \underline{\underline{A}}_{\Delta\Pi}\left(\underline{\underline{A}}_{\Pi\Pi}\right)^{-1}\underline{\underline{A}}_{\Pi\Delta}$$

$$\underline{vec}_{\Delta} \leftarrow \underline{\underline{S}}\,\underline{u}_{\Delta}$$

$$\underline{\underline{A}}_{\Delta\Delta} = \underline{\underline{A}}^{t}_{\Delta\Delta}; \quad \underline{\underline{A}}^{t}_{\Delta\Delta} = \sum_{\alpha=1}^{E} \underline{\underline{A}}^{t,\alpha}_{\Delta\Delta}$$

$$\underline{\underline{A}} = \underline{\underline{a}}'\,\underline{\underline{A}}^{t}\,\underline{\underline{a}}'$$

$$\underline{\underline{A}}_{\Pi\Delta} = \underline{\underline{a}}'\,\underline{\underline{A}}^{t}_{\Pi\Delta} = \underline{\underline{a}}'\left(\sum_{\alpha=1}^{E} \underline{\underline{A}}^{t,\alpha}_{\Pi\Delta}\right)$$

$$\underline{\underline{A}}_{\Delta\Pi} = \underline{\underline{A}}^{t}_{\Delta\Pi}\,\underline{\underline{a}}' = \left(\sum_{\alpha=1}^{E} \underline{\underline{A}}^{t,\alpha}_{\Delta\Pi}\right)\underline{\underline{a}}'$$

Schur complement: $\underline{vec}_\Delta \leftarrow \underline{\underline{S}}\,\underline{u}_\Delta$

Sub-sets: $M = \Pi = I \cup \pi$ and $N = \Delta$

domain06.png

$$\boxed{\begin{pmatrix} \underline{\underline{A}}_{\Pi\Pi} & \underline{\underline{A}}_{\Pi\Delta} \\ \underline{\underline{A}}_{\Delta\Pi} & \underline{\underline{A}}_{\Delta\Delta} \end{pmatrix} \begin{pmatrix} \underline{u}_\Pi \\ \underline{u}_\Delta \end{pmatrix} = \begin{pmatrix} \underline{f}_\Pi \\ \underline{f}_\Delta \end{pmatrix}}$$

- Schur Complement Formulation

Apply: $\boxed{\underline{\underline{S}}\,\underline{u}_\Delta = \sigma_{\Delta\Delta}(\underline{\underline{A}})\underline{u}_\Delta}$

where $\sigma_{\Delta\Delta}(\underline{\underline{A}}) = \underline{\underline{A}}_{\Delta\Delta} - \underline{\underline{A}}_{\Delta\Pi}(\underline{\underline{A}}_{\Pi\Pi})^{-1}\underline{\underline{A}}_{\Pi\Delta}$

$$\underline{\underline{A}}_{\Pi\Delta}\underline{u}_\Delta = \underline{\mathrm{v}}_\Pi$$

$$(\underline{\underline{A}}_{\Pi\Pi})^{-1}\underline{\mathrm{v}}_\Pi = \underline{\mathrm{w}}_\Pi \rightarrow \underline{\underline{A}}_{\Pi\Pi}\underline{\mathrm{w}}_\Pi = \underline{\mathrm{v}}_\Pi$$

Application of $(\underline{\underline{A}}_{\Pi\Pi})^{-1}\underline{\mathrm{v}}_{\Pi} = \underline{\mathrm{w}}_{\Pi}$

Sub-sets: $M = \Pi = I$ and $N = \pi$

domain07.png

$$\begin{pmatrix} \underline{\underline{A}}_{II} & \underline{\underline{A}}_{I\pi} \\ \underline{\underline{A}}_{\pi I} & \underline{\underline{A}}_{\pi\pi} \end{pmatrix} \begin{pmatrix} \underline{\mathrm{w}}_{I} \\ \underline{\mathrm{w}}_{\pi} \end{pmatrix} \begin{pmatrix} \underline{\mathrm{v}}_{I} \\ \underline{\mathrm{v}}_{\pi} \end{pmatrix}$$

- Schur Complement Formulation

Solve: $\boxed{\sigma_{\pi\pi}(\underline{\underline{A}}_{\Pi\Pi})\underline{\mathrm{w}}_{\pi} = \underline{g}_{\pi}}$

where $\sigma_{\pi\pi}(\underline{\underline{A}}_{\Pi\Pi}) = \underline{\underline{a}}' \left(\underline{\underline{A}}^{t}_{\pi\pi} - \underline{\underline{A}}^{t}_{\pi I}(\underline{\underline{A}}^{t}_{II})^{-1}\underline{\underline{A}}^{t}_{I\pi}\right)\underline{\underline{a}}'$

and $\underline{g}_{\pi} = \underline{\mathrm{v}}_{\pi} - \underline{\underline{a}}'\,\underline{\underline{A}}^{t}_{\pi I}(\underline{\underline{A}}^{t}_{II})^{-1}\underline{\mathrm{v}}_{I}$

Calculate: $\boxed{\underline{\mathrm{w}}_{I} = (\underline{\underline{A}}^{t}_{II})^{-1}(\underline{\mathrm{v}}_{I} - \underline{\underline{A}}_{I\pi}\underline{\mathrm{w}}_{\pi})}$

$\rightarrow \underline{\mathrm{w}}_{\Pi} = \underline{\mathrm{w}}_{I} + \underline{\mathrm{w}}_{\pi}$

Application of $\underline{\underline{S}}^{-1}$

Basic result:

$$(\underline{\underline{A}}^{-1})_{\Delta\Delta}\underline{w} = \left(\underline{\underline{A}}^{-1}\underline{w}_{\Delta}\right)_{\Delta} = \underline{\underline{S}}^{-1}\underline{w}_{\Delta}$$

Define: $\Sigma \equiv I \cup \Delta$

Then: $X = \Sigma \cup \pi$ and $\varnothing = \Sigma \cap \pi$

Furthemore:

$$\underline{\underline{A}} = \begin{pmatrix} \underline{\underline{A}}_{\Sigma\Sigma} & \underline{\underline{A}}_{\Sigma\pi} \\ \underline{\underline{A}}_{\pi\Sigma} & \underline{\underline{A}}_{\pi\pi} \end{pmatrix} = \begin{pmatrix} \underline{\underline{A}}^{t}_{\Sigma\Sigma} & \underline{\underline{A}}^{t}_{\Sigma\pi}\underline{\underline{a}}' \\ \underline{\underline{a}}'\underline{\underline{A}}^{t}_{\pi\Sigma} & \underline{\underline{a}}'\underline{\underline{A}}^{t}_{\pi\pi}\underline{\underline{a}}' \end{pmatrix}$$

$$\underline{\underline{A}}^{-1} = \begin{pmatrix} (\underline{\underline{A}}^{-1})_{\Pi\Pi} & (\underline{\underline{A}}^{-1})_{\Pi\Delta} \\ (\underline{\underline{A}}^{-1})_{\Delta\Pi} & (\underline{\underline{A}}^{-1})_{\Delta\Delta} \end{pmatrix} = \begin{pmatrix} (\underline{\underline{A}}^{-1})_{\Pi\Pi} & (\underline{\underline{A}}^{-1})_{\Pi\Delta} \\ (\underline{\underline{A}}^{-1})_{\Delta\Pi} & \underline{\underline{S}}^{-1} \end{pmatrix}$$

Inverse of Schur $\underline{vec}_\Delta \leftarrow \underline{\underline{S}}^{-1} \underline{f}_\Delta$

Because $\underline{\underline{S}}^{-1} = (\underline{\underline{A}}^{-1})_{\Delta\Delta}$ then
solve $\underline{\underline{A}}\,\underline{u} = \underline{f}$ and restrict $\underline{v}$ to Δ
Sub-sets: $M = \Sigma = I \cup \Delta$ and $N = \pi$

domain08.png

$$\boxed{\begin{pmatrix} \underline{\underline{A}}^t_{\Sigma\Sigma} & \underline{\underline{A}}^t_{\Sigma\pi}\,\underline{\underline{a}}' \\ \underline{\underline{a}}'\,\underline{\underline{A}}^t_{\pi\Sigma} & \underline{\underline{a}}'\,\underline{\underline{A}}^t_{\pi\pi}\,\underline{\underline{a}}' \end{pmatrix} \begin{pmatrix} \underline{u}_\Sigma \\ \underline{u}_\pi \end{pmatrix} = \begin{pmatrix} \underline{f}_\Sigma \\ \underline{f}_\pi \end{pmatrix}}$$

- Schur Complement Formulation

Solve: $\boxed{\sigma_{\pi\pi}(\underline{\underline{A}})\underline{u}_\pi = \underline{g}_\pi}$

where $\sigma_{\pi\pi}(\underline{\underline{A}}) = \underline{\underline{a}}' \left(\underline{\underline{A}}^t_{\pi\pi} - \underline{\underline{A}}^t_{\pi\Sigma} (\underline{\underline{A}}^t_{\Sigma\Sigma})^{-1} \underline{\underline{A}}^t_{\Sigma\pi} \right) \underline{\underline{a}}'$

and $\underline{g}_\pi = \underline{f}_\pi - \underline{\underline{a}}'\underline{\underline{A}}^t_{\pi\Sigma} (\underline{\underline{A}}^t_{\Sigma\Sigma})^{-1} \underline{f}_\Sigma$

Calculate: $\boxed{\underline{u}_\Sigma = (\underline{\underline{A}}^t_{\Sigma\Sigma})^{-1} (\underline{f}_\Sigma - \underline{\underline{A}}^t_{\Sigma\pi}\,\underline{u}_\pi)}$

$$\underline{u} = \underline{u}_\Sigma + \underline{u}_\pi = \underline{u}_I + \underline{u}_\Delta + \underline{u}_\pi$$

$$\rightarrow \boxed{\underline{\underline{S}}^{-1} \underline{f}_\Delta = \underline{u}_\Delta}$$

Finite Volume Method: partition

- We make the partition on the centers of the volumes.

meshFVMPart.pdf

Finite Volume Method: subdomains

meshFVMSubdom.pdf

meshFVMSubdom1.pdf

i		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
	j	1				2				3				4			
1	1	P	E			N											
2	1	W	P	E			N										
3	1		W	P	E			N									
1	2	S				P	E			N							
2	2		S			W	P	E			N						
3	2			S			W	P	E			N					
1	3					S				P	E			N			
2	3						S			W	P	E			N		
3	3							S			W	P	E			N	
1	4									S				P	E		
2	4										S			W	P	E	
3	4											S			W	P	E

meshFVMSubdom2.pdf

i		2	3	4	5	2	3	4	5	2	3	4	5	2	3	4	5
	j	1				2				3				4			
3	1	W	P	E			N										
4	1		W	P	E			N									
5	1			W	P				N								
3	2		S			W	P	E			N						
4	2			S			W	P	E			N					
5	2				S			W	P				N				
3	3						S			W	P	E			N		
4	3							S			W	P	E			N	
5	3								S			W	P				N
3	4										S			W	P	E	
4	4											S			W	P	E
5	4												S			W	P

meshFVMSubdomHalo.pdf

TUNAM

code.google.com/p/tunam

Meshes in TUNAM

meshFVMPart.pdf

- Empty circles: Nodes on the real boundary.
- Filled circles: Nodes in the interior of the domain.
- The nodes on the real boundary are treated as volumes, in such a way that all coefficients are calculated on all these points.

Subdomains in TUNAM

SubDomainDVS_TUNAM.pdf

TUNAM: Storage (Subdomain 1)

SubDomain_1_DVS_TUNAM.p

i	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
	1				2				3				4			
π	***P***	***E***			***N***											
Δ	***W***	***P***	***E***			***N***										
π		***W***	***P***	***E***			***N***									
Δ	***S***				***P***	***E***			***N***							
I		***S***			***W***	***P***	***E***			***N***						
Δ			***S***			***W***	***P***	***E***			***N***					
Δ					***S***				***P***	***E***			***N***			
I						***S***			***W***	***P***	***E***			***N***		
Δ							***S***			***W***	***P***	***E***			***N***	
π									***S***				***P***	***E***		
Δ										***S***			***W***	***P***	***E***	
π											***S***			***W***	***P***	***E***

DP	π	Δ	π	Δ	I	Δ	Δ	I	Δ	π	Δ	π
i	1	2	3	1	2	3	1	2	3	1	2	3
j	1	1	1	2	2	2	3	3	3	4	4	4
store[0] $\rightarrow$	***P***	***P***	***P***	***P***	***P***	***P***	***P***	***P***	***P***	***P***	***P***	***P***
store[1] $\rightarrow$	***E***	***E***	***E***	***E***	***E***	***E***	***E***	***E***	***E***	***E***	***E***	***E***
store[2] $\rightarrow$	***W***	***W***	***W***	***W***	***W***	***W***	***W***	***W***	***W***	***W***	***W***	***W***
store[3] $\rightarrow$	***N***	***N***	***N***	***N***	***N***	***N***	***N***	***N***	***N***	***N***	***N***	***N***
store[4] $\rightarrow$	***S***	***S***	***S***	***S***	***S***	***S***	***S***	***S***	***S***	***S***	***S***	***S***
	0	1	2	3	4	5	6	7	8	9	10	11

Mat-Vec Mult

SubDomain_1_DVS UNAM.pdf

	i		1	2	3	1	2	3	1	2	3	1	2	3	DP
		j	1	1	1	2	2	2	3	3	3	4	4	4	
0	1	1	P	E		N									π
1	2	1	W	P	E		N								Δ
2	3	1		W	P			N							π
3	1	2	S			P	E		N						Δ
4	2	2		S		W	P	E		N					I
5	3	2			S		W	P			N				Δ
6	1	3				S			P	E		N			Δ
7	2	3					S		W	P	E		N		I
8	3	3						S		W	P			N	Δ
9	1	4							S			P	E		π
10	2	4								S		W	P	E	Δ
11	3	4									S		W	P	π
			0	1	2	3	4	5	6	7	8	9	10	11	

	0										1										2				
	0	1	2	**3**	4	5	6	**7**	8	9	**0**	1	2	3	**4**	5	6	7	8	**9**	0	1	2	. .
val	P	E	N	W	P	E	N	W	P	N	S	P	E	N	S	W	P	E	N	S	W	P	N	. .
col	0	1	3	0	1	2	4	1	2	5	0	3	4	6	1	3	4	5	7	2	4	5	8	. .

row_ptr	0	3	7	10	14	19	. . .	**46**
								nnz

TUNAM: example 1D

```
typedef float prec_t;
typedef TunaArray<prec_t,1>::huge ScalarField1D;

StructuredMesh<Uniform<prec_t, 1> > mesh(length_x, num_nodes_x);
mesh.print();
prec_t dx = mesh.getDelta(X);

ScalarField1D T (mesh.getExtentVolumes());
ScalarField1D Tn(mesh.getExtentNodes());

firstIndex i;
Tn = 10 * cos (3 * PI * dx * i);
NumUtils::interpolateToVolumes(T, Tn);
InOut::writeToFile(Tn, 0, "./Data1D/temp.", dx);
```

Output:

```
+----- TUNA info -----+
| StructuredMesh<  Uniform < 1D > >
+-----+
| X-axis : length = 1; nodes = 4; volumes = 5; delta = 0.333333
+---------------------+
```

TUNAM: exmaple 1D

```
SparseMatrix< Diagonal<prec_t, 1> > A(num_nodes_x);
ScalarField1D                       b(num_nodes_x);

ScalarEquation < LaplaceScheme<prec_t, 1> > laplace(T, A, b, mesh.getDeltas());
laplace.setGamma(1.0);
laplace.setDirichlet(LEFT_WALL);
laplace.setDirichlet(RIGHT_WALL);
laplace.print();
laplace.calcCoefficients();
```

Ouput:

```
+----- TUNA info -----+
| GeneralEquation< ScalarEquation  >
+-----+
| bi = 1  ei = 3
| bj = 0  ej = 1
| bk = 0  ek = 1
| dx = 0.333333  dy = 1
| dz = 1  dt = 1
+-----+
| Boundary conditions
+-----+
| Dirichlet: Wall = LEFT
| Dirichlet: Wall = RIGHT
+-----+
| ScalarEquation<  LaplaceScheme < 1D  > >
+-----+
| Gamma = 1
+--------------------+
```

TUNAM: example 1D

```
cout << "\n ----- STORING WITH TUNAM ----- \n";
cout << " Initital T = " << T << endl;
cout << "\n aP = " << laplace.aP << endl;
cout << "\n aE = " << laplace.aE << endl;
cout << "\n aW = " << laplace.aW << endl;
cout << "\n b = " << b << endl;
```

Output:

```
----- STORING WITH TUNAM -----
Initital T = 5
[        10         0         0         0       -10  ]

aP = 4
[         0         9         6         9  ]

aE = 4
[         0         3         3         0  ]

aW = 4
[         0         0         3         3  ]

b = 4
[         0        60         0       -60  ]
```

TUNAM: example 1D

```
extern int csr_from_cusp(int *, int *, prec_t *, prec_t *, prec_t *, int, int);

void csr_1D(SparseMatrix< Diagonal<prec_t, 1> > &A, ScalarField1D &b, ScalarField1D & T, int↩
      n);

csr_1D(A, b, T, num_nodes_x-1);
```

Output:

```
----- CSR FORMAT IN TUNAM -----
val = 9  3  3  6  3  3  9
col = 0  1  0  1  2  1  2
row = 0  2  5  7

b[0] = 60  T[0] = 0
b[1] = 0  T[1] = 0
b[2] = -60  T[2] = 0
```

```
----- CSR FORMAT IN CUSP -----
sparse matrix <3, 3> with 7 entries
              0              0              9
              0              1              3
              1              0              3
              1              1              6
              1              2              3
              2              1              3
              2              2              9
 b[0] = 60  x[0] = 0
 b[1] = 0  x[1] = 0
 b[2] = -60  x[2] = 0
```

TUNAM: interface to CUSP

```
//typedef cusp::device_memory MemorySpace;
typedef cusp::host_memory MemorySpace;
typedef float ValueType;

int csr_from_cusp(int *h_rp, int *h_cj, float *h_va, float *h_b, float *h_T, int n, int nnz)
{
  // allocate storage for (n,n) matrix with nnz nonzeros in csr format
  cusp::csr_matrix<int,ValueType,MemorySpace> A(n,n,nnz);

  // initialize row ptr array
  for(int i = 0; i < n+1; i++)
    A.row_offsets[i] = h_rp[i];

  // initialize column and value arrays
  for(int i = 0; i < nnz; i++) {
    A.column_indices[i] = h_cj[i];
    A.values[i] = h_va[i];
  }

  // print matrix entries
  cout << "\n ----- CSR FORMAT IN CUSP ----- \n";
  cusp::print(A);

  // allocate storage for solution (x) and right hand side (b)
  cusp::array1d<ValueType, MemorySpace> x(A.num_rows, 0);
  cusp::array1d<ValueType, MemorySpace> b(A.num_rows, 0);

  return 0;
}
```

Mat-Vec Mult

SubDomain_1_DVS_TUNAM.pdf

i		1	2	3	1	2	3	1	2	3	1	2	3	DP
	j	1	1	1	2	2	2	3	3	3	4	4	4	
1	1	P	E		N									π
2	1	W	P	E		N								Δ
3	1		W	P			N							π
1	2	S			P	E		N						Δ
2	2		S		W	P	E		N					I
3	2			S		W	P			N				Δ
1	3				S			P	E		N			Δ
2	3					S		W	P	E		N		I
3	3						S		W	P			N	Δ
1	4							S			P	E		π
2	4								S		W	P	E	Δ
3	4									S		W	P	π

$$\begin{pmatrix} \underline{\underline{A}}_{II} & \underline{\underline{A}}_{I\Delta} & \underline{\underline{A}}_{I\pi} \\ \underline{\underline{A}}_{\Delta I} & \underline{\underline{A}}_{\Delta\Delta} & \underline{\underline{A}}_{\Delta\pi} \\ \underline{\underline{A}}_{\pi I} & \underline{\underline{A}}_{\pi\Delta} & \underline{\underline{A}}_{\pi\pi} \end{pmatrix} =$$

i		2	2	2	1	3	1	3	2	1	3	1	3	DP
	j	2	3	1	2	2	3	3	4	1	1	4	4	
2	2	P	N	S	W	E								I
2	3	S	P				W	E	N					I
2	1	N		P						W	E			Δ
1	2	E			P		N			S				Δ
3	2	W				P		N			S			Δ
1	3		E		S		P					N		Δ
3	3		W			S		P					N	Δ
2	4		S						P			W	E	Δ
1	1			E	N					P				π
3	1			W		N					P			π
1	4						S		E			P		π
3	4							S	W				P	π

Mat-Vec: $\underline{\underline{A}}_{\Sigma\Sigma} \times \underline{u}_{\Sigma}$

$\underline{\underline{A}}_{\Sigma\Sigma} \times \underline{u}_{\Sigma} =$

i		2	2	2	1	3	1	3	2	DP
	j	2	3	1	2	2	3	3	4	
2	2	P	N	S	W	E				I
2	3	S	P				W	E	N	I
2	1	N		P						Δ
1	2	E			P		N			Δ
3	2	W				P		N		Δ
1	3		E		S		P			Δ
3	3		W			S		P		Δ
2	4		S						P	Δ

$\times$

1
1
1
1
1
1
1
1

$=$

$P+N+S+W+E$
$S+P+W+E+N$
$N+P$
$E+P+N$
$W+P+N$
$E+S+P$
$W+S+P$
$S+P$

i		1	2	3	1	2	3	1	2	3	1	2	3	DP
	j	1	1	1	2	2	2	3	3	3	4	4	4	
1	1	P	E		N									$\pi = 0$
2	1	W	P	E		N								Δ
3	1		W	P			N							$\pi = 0$
1	2	S			P	E		N						Δ
2	2		S		W	P	E		N					I
3	2			S		W	P			N				Δ
1	3				S			P	E		N			Δ
2	3					S		W	P	E		N		I
3	3						S		W	P			N	Δ
1	4							S			P	E		$\pi = 0$
2	4								S		W	P	E	Δ
3	4									S		W	P	$\pi = 0$

$=$

$P+N$
$P+E+N$
$S+W+P+E+N$
$W+P+N$
$S+P+E$
$S+W+P+E+N$
$S+W+P$
$S+P$

Virtual topology

topovirtual.png

SubDom.jpg

Application of average requires MPI communication

comm00.pdf

- Initial configuration.

Average application. Communication : stage 1

comm01.pdf

- Send RIGHT, Recv LEFT.

Average application. Communication : stage 2

comm02.pdf

- Send RIGHT, Recv LEFT.
- Send LEFT, Recv RIGHT.

Average application. Communication : stage 3

comm03.pdf

- Send RIGHT, Recv LEFT.
- Send LEFT, Recv RIGHT.
- Send TOP, Recv DOWN.

Average application. Communication : stage 4

comm04.pdf

- Send RIGHT, Recv LEFT.
- Send LEFT, Recv RIGHT.
- Send TOP, Recv DOWN.
- Send DOWN, Recv TOP.

Average application

comm04.pdf

- Send RIGHT, Recv LEFT.
- Send LEFT, Recv RIGHT.
- Send TOP, Recv DOWN.
- Send DOWN, Recv TOP.
- Apply $\underline{\underline{a}}$ on all walls.